

Machine Learning — Linear Regression

Numerical Questions with Detailed Step-by-Step Solutions

Clean mathematical notation • Matrices • Closed-Form Solution • Gradient Descent • MSE

Simple Level

Question 1

Given the data points (1, 2), (2, 4), and (3, 6), calculate the slope m of the regression line.

Step-by-Step Solution

Step 1: Calculate the required sums.

$$n = 3, \Sigma x = 6, \Sigma y = 12, \Sigma xy = 28, \Sigma x^2 = 14$$

Step 2: Use the slope formula.

$$m = [n\Sigma xy - (\Sigma x)(\Sigma y)] / [n\Sigma x^2 - (\Sigma x)^2]$$

Step 3: Substitute the values.

$$m = [3(28) - 6(12)] / [3(14) - 6^2]$$

Step 4: Simplify.

$$m = (84 - 72) / (42 - 36) = 12 / 6 = 2$$

Final Answer: $m = 2$

Simple Level

Question 2

Using the same data points (1, 2), (2, 4), and (3, 6), calculate the intercept b.

Step-by-Step Solution

Step 1: Calculate the means.

$$\bar{x} = 2, \bar{y} = 4$$

Step 2: Use $b = \bar{y} - m\bar{x}$.

$$b = 4 - 2(2) = 0$$

Final Answer: $b = 0$. Regression line: $\hat{y} = 2x$.

Simple Level

Question 3

For the linear regression model below, calculate the predicted value when $x = 5$.

$$\hat{y} = 2x + 1$$

Step-by-Step Solution

Step 1: Write the model.

$$\hat{y} = 2x + 1$$

Step 2: Substitute $x = 5$.

$$\hat{y} = 2(5) + 1$$

Step 3: Calculate.

$$\hat{y} = 10 + 1 = 11$$

Final Answer: $\hat{y} = 11$

Simple Level

Question 4

Actual values are $y = [3, 5, 7]$ and predicted values are $\hat{y} = [2, 6, 8]$. Calculate the Mean Squared Error (MSE).

$$\text{MSE} = (1/n) \sum (y_i - \hat{y}_i)^2$$

Step-by-Step Solution

Step 1: Calculate the errors.

$$1, -1, -1$$

Step 2: Square the errors.

$$1^2 = 1, (-1)^2 = 1, (-1)^2 = 1$$

Step 3: Calculate MSE.

$$\text{MSE} = (1 + 1 + 1) / 3 = 1$$

Final Answer: MSE = 1

Medium Level

Question 5

Given $x = [1, 2, 3, 4]$ and $y = [3, 5, 7, 9]$, calculate the regression equation $\hat{y} = mx + b$.

Step-by-Step Solution

Step 1: Calculate the means.

$$\bar{x} = 2.5, \bar{y} = 6.5$$

Step 2: Calculate the slope.

$$m = 2$$

Step 3: Calculate the intercept.

$$b = \bar{y} - m\bar{x} = 6.5 - 2(2.5) = 1$$

Step 4: Write the regression equation.

$$\hat{y} = 2x + 1$$

Final Answer: $\hat{y} = 2x + 1$

Medium Level

Question 6

Given the feature matrix X and target vector y below, calculate θ using the closed-form normal equation.

Feature matrix X :

1	1
1	2
1	3

Target vector y :

2
4
6

$$\theta = (X^T X)^{-1} X^T y$$

Step-by-Step Solution

Step 1: Calculate X^T .

1	1	1
1	2	3

Step 2: Calculate $X^T X$.

3	6
6	14

Step 3: Calculate $X^T y$.

12
28

Step 4: Set up the normal equations.

$$3b + 6m = 12 \quad 6b + 14m = 28$$

Step 5: Solve.

$$b = 4 - 2m \rightarrow m = 2 \rightarrow b = 0$$

Final Answer: $\theta = [0, 2]^T$; therefore $b = 0$ and $m = 2$.

Medium Level

Question 7

For the model $\hat{y} = 2x + 1$ and data points (1, 4), (2, 5), and (3, 8), calculate the MSE.

$$\text{MSE} = (1/n) \sum (y_i - \hat{y}_i)^2$$

Step-by-Step Solution

Step 1: Calculate predictions.

$$\hat{y} = [3, 5, 7]$$

Step 2: Calculate squared errors.

$$(4-3)^2 = 1, (5-5)^2 = 0, (8-7)^2 = 1$$

Step 3: Calculate MSE.

$$\text{MSE} = (1 + 0 + 1) / 3 = 2/3 \approx 0.667$$

Final Answer: MSE $\approx$ 0.667

Medium Level

Question 8

Perform one gradient descent update using $x = 2$, $y = 6$, initial $w = 1$, $b = 0$, and learning rate $\alpha = 0.1$. Use the cost function shown below.

$$\hat{y} = wx + b$$

$$J = \frac{1}{2}(\hat{y} - y)^2$$

$$\partial J / \partial w = (\hat{y} - y)x \quad \partial J / \partial b = \hat{y} - y$$

Step-by-Step Solution

Step 1: Calculate prediction.

$$\hat{y} = 1(2) + 0 = 2$$

Step 2: Calculate error.

$$\hat{y} - y = 2 - 6 = -4$$

Step 3: Calculate gradients.

$$\partial J / \partial w = (-4)(2) = -8$$

$$\partial J / \partial b = -4$$

Step 4: Update w .

$$w_{\text{new}} = 1 - 0.1(-8) = 1.8$$

Step 5: Update b .

$$b_{\text{new}} = 0 - 0.1(-4) = 0.4$$

Final Answer: $w = 1.8$ and $b = 0.4$

Hard Level

Question 9

Using batch gradient descent, perform one iteration for data points (1, 3), (2, 5), and (3, 7). Start with $w = 0$, $b = 0$ and learning rate $\alpha = 0.1$. Calculate the updated w and b .

$$J = (1/2m) \sum (\hat{y}_i - y_i)^2$$

Step-by-Step Solution

Step 1: Initial predictions.

$$\hat{y}_1 = \hat{y}_2 = \hat{y}_3 = 0$$

Step 2: Errors.

$$e = [-3, -5, -7]$$

Step 3: Gradient for w .

$$\partial J / \partial w = (1/3)[(-3)(1) + (-5)(2) + (-7)(3)] = -10.667$$

Step 4: Gradient for b .

$$\partial J / \partial b = (-3 - 5 - 7) / 3 = -5$$

Step 5: Update w .

$$w_{new} = 0 - 0.1(-10.667) \approx 1.067$$

Step 6: Update b .

$$b_{new} = 0 - 0.1(-5) = 0.5$$

Final Answer: $w \approx 1.067$ and $b = 0.5$

Hard Level

Question 10

For $x = [1, 2, 3]$ and $y = [2, 4, 5]$, calculate the closed-form slope and intercept, then predict y when $x = 4$.

$$\theta = (X^T X)^{-1} X^T y$$

Step-by-Step Solution

Step 1: Calculate the means.

$$\bar{x} = 2, \bar{y} = 11/3$$

Step 2: Calculate the slope.

$$m = 1.5$$

Step 3: Calculate the intercept.

$$b = 11/3 - 1.5(2) = 2/3$$

Step 4: Write the equation.

$$\hat{y} = 1.5x + 2/3$$

Step 5: Predict at $x = 4$.

$$\hat{y}(4) = 1.5(4) + 2/3 = 20/3 \approx 6.667$$

Final Answer: $m = 1.5$, $b = 2/3$, $\hat{y}(4) \approx 6.667$

Hard Level

Question 11

Given the feature matrix X and target vector y below, calculate θ using the closed-form solution.

Feature matrix X :

1	1
1	2
1	4

Target vector y :

2
3
7

$$\theta = (X^T X)^{-1} X^T y$$

Step-by-Step Solution

Step 1: Calculate X^T .

1	1	1
1	2	4

Step 2: Calculate $X^T X$.

3	7
7	21

Step 3: Calculate $X^T y$.

12
36

Step 4: Calculate determinant.

$$\det(X^T X) = 3(21) - 7(7) = 14$$

Step 5: Calculate the inverse.

21/14	-7/14
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$-7/14$	$3/14$
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Step 6: Multiply by $X^T y$.

21	-7
-7	3

Step 7: Calculate the first parameter.

$$\theta_1 = [21(12) - 7(36)] / 14 = 0$$

Step 8: Calculate the second parameter.

$$\theta_2 = [-7(12) + 3(36)] / 14 = 12/7$$

Final Answer: $\theta = [0, 12/7]^T$. Therefore $b = 0$ and $m = 12/7 \approx 1.714$.

Hard Level

Question 12

A model starts with $w = 1$ and $b = 1$. For $x = 3$ and $y = 10$, perform one gradient descent step with learning rate $\alpha = 0.05$. Find the updated w and b .

$$\hat{y} = wx + b$$

$$J = \frac{1}{2}(\hat{y} - y)^2$$

$$\partial J / \partial w = (\hat{y} - y)x \quad \partial J / \partial b = \hat{y} - y$$

Step-by-Step Solution

Step 1: Calculate prediction.

$$\hat{y} = 1(3) + 1 = 4$$

Step 2: Calculate error.

$$\hat{y} - y = 4 - 10 = -6$$

Step 3: Calculate gradients.

$$\partial J / \partial w = (-6)(3) = -18$$

$$\partial J / \partial b = -6$$

Step 4: Update w .

$$w_{\text{new}} = 1 - 0.05(-18) = 1.9$$

Step 5: Update b .

$$b_{\text{new}} = 1 - 0.05(-6) = 1.3$$

Final Answer: $w = 1.9$ and $b = 1.3$